Wenkai Qin

CONTACT INFORMATION

Department of Aerospace Engineering and Engineering Mechanics
The University of Texas at Austin
2617 Wichita Street, C0600
Austin, TX 78712
Email: todd.humphreys@utexas.edu *Office:* (512) 471-4489

EDUCATION

Cornell University, Ithaca, New York USA

Ph.D., Aerospace Engineering, January 2008
Advisors: Mark L. Psiaki (primary) and Paul M. Kintner (co-advisor)
Dissertation: Modeling Ionospheric Scintillation and its Effects on GPS Carrier Tracking Loops and Two Other Applications of Modeling and Estimation

Utah State University, Logan, Utah USA

M.S., Electrical and Computer Engineering, May 2003
Thesis: Attitude Determination For Small Satellites With Modest Pointing Constraints

Utah State University, Logan, Utah USA

B.S., Electrical and Computer Engineering, May 2000
Minors: Mathematics and Spanish

CURRENT AND PREVIOUS ACADEMIC POSITIONS

Professor, The University of Texas at Austin — September 2021 - present
Department of Aerospace Engineering and Engineering Mechanics
Director of the Radionavigation Laboratory
Director of the Wireless Networking and Communications Group

Associate Professor, The University of Texas at Austin — September 2015 - August 2021
Department of Aerospace Engineering and Engineering Mechanics
Director of the Radionavigation Laboratory

Assistant Professor, The University of Texas at Austin — August 2009- August 2015
Department of Aerospace Engineering and Engineering Mechanics
Director of the Radionavigation Laboratory

Adjunct Faculty, Cornell University — August 2007 - January 2008
Instructor for graduate-level course in model-based estimation.

Post-Doctoral Researcher, Cornell University — August 2007 - August 2008

Research Assistant, Cornell University — February 2003 - July 2007

OTHER PROFESSIONAL EXPERIENCE

Co-Founder, Radiosense LLC — February 2015 - February 2017
Centimeter-accurate positioning for the mass market.

Co-Founder, Coherent Navigation Inc. August 2008 - August 2009
Exploited Iridium constellation to harden navigation and timing sensors. Acquired by Apple.

IONF Satellite Constellation Attitude Determination Lead, Space Dynamics Laboratory 2001 - 2002

MEMS Inertial Sensors Researcher, NASA Jet Propulsion Laboratory 1999 - 2002 (summers)

MEMBERSHIPS IN PROFESSIONAL AND HONORARY SOCIETIES

Fellow, Institute of Navigation (ION)
Fellow, The Royal Institute of Navigation (RIN)
Member, Institute of Electrical and Electronics Engineers (IEEE)

COMMITTEES, BOARDS, AND CONFERENCES ORGANIZED/CHAIRED

Outside Committees

Land Representative, Institute of Navigation Governing Council, 2012–2014

Conference Activities: Organizer

Track chair, European Space Agency NAVITEC conference, April 2022

Lead organizer, Texas Wireless Summit, November 2016

Track chair, IEEE/Institute of Navigation PLANS conference, April 2016

Lead organizer, Texas Wireless Summit, October 2013

Track chair, Institute of Navigation GNSS+ conference, September 2013

Lead organizer, Civil GPS Security Meeting, September 2010

Conference Activities: Session Chair

Panel session chair, Brooklyn 6G Summit: The Metaverse, October 2022

Session chair, Institute of Navigation GNSS+ conference: Extended Reality and PNT, September 2022

Session chair, 6G@UT Forum, Extraterrestrial 6G, May 2022

Session chair, IEEE Wireless Communications and Networking Conference: Co-Design of Signals for Sensing and Communications, April 2022

Session chair, Munich Satellite Summit, Jamming and Spoofing in GNSS, March 2022

Session chair, Institute of Navigation GNSS+ conference: GNSS Authentication and Anti-Spoofing, September 2021

Session chair, Institute of Navigation GNSS+ conference: Novel Applications of GNSS in Smartphones, September 2020

Session chair, IEEE/ION PLANSx conference: Autonomous Vehicle Navigation in Challenging Environments, September 2020

Session chair, IEEE/ION PLANS conference: GNSS Resilience, May 2018

Session chair, Institute of Navigation GNSS conference: High Precision GNSS Positioning, September 2016

Session chair, Institute of Navigation GNSS conference: GNSS Vulnerabilities and Anti-Jamming, September 2015

Panel Session chair, Institute of Navigation GNSS conference: Privacy Issues, September 2015

Session chair, Institute of Navigation GNSS conference: GNSS Vulnerabilities 1: Interference, September 2014

Session chair, Institute of Navigation GNSS conference: GNSS Vulnerabilities 2: Spoofing and Authentication, September 2014

Panel Session chair, Institute of Navigation GNSS conference: GNSS Vulnerabilities and Threats, September 2014

Session chair, IEEE/ION PLANS conference: Interference and Robust Navigation, May 2014

Session chair, Institute of Navigation International Technical Meeting: Interference and Spectrum Management, January 2013

Session chair, Institute of Navigation GNSS conference: Spectrum Interference, September 2012

Session chair, IEEE/ION PLANS conference: Receiver and Antenna Technology Session, April 2012

Panel session chair, Institute of Navigation GNSS conference: GNSS Security, September 2011

Session chair, IEEE/ION PLANS conference: Algorithms and Processing, May 2010

Journal Activities
Editor, IEEE Transactions on Wireless Communications, May 2014–January 2017

OTHER PROFESSIONAL HIGHLIGHTS

Congressional Testimony
Invited witness, U.S. House Subcommittee on Homeland Security Oversight hearing on the threat of unmanned aerial systems, March 2015

Invited witness, U.S. House Judiciary Subcommittee on Crime, Terrorism, and Homeland Security field forum on privacy in an age of drones, October 2012

Invited witness, U.S. House Subcommittee on Homeland Security Oversight hearing on drone security, July 2012

Invited Subject Matter Expert
National Executive Committee on Position Navigation and Timing, Advisory Board Meeting, June 2019

National Executive Committee on Position Navigation and Timing, Advisory Board Meeting, June 2015

Department of Homeland Security risk assessment of critical infrastructure dependence on GPS, March-September, 2011

U.S. Patent and Trademark Office briefing on advances in GNSS technology, April 2011

National Executive Committee on Position Navigation and Timing, Advisory Board Meeting, October 2010

Current Review Activities
IEEE Transactions on Aerospace and Electronic Systems

NAVIGATON, the journal of the Institute of Navigation

IEEE Transactions on Wireless Communications

UNIVERSITY COMMITTEES/ADMINISTRATIVE ASSIGNMENTS

Administrative Assignments
Director, Wireless Networking and Communications Group, 2021-present

Director, Radionavigation Laboratory, 2009-present

Director, UT SAVES, 2020-2021

Associate Director, UT SAVES, 2016-2020

Faculty advisor, UT student chapter of AIAA, 2010-2019

Faculty advisor, Sigma Gamma Tau Honor Society, 2016-2019

Cockrell School of Engineering
Cockrell School Honors Committee, 2011-2019

College of Engineering Faculty Committee

Department Committees
ECE Faculty Search Committee, 2021-2022

ASE Chair of Robotics Faculty Search Committee, 2019-2020

ASE Strategic Planning Committee, 2017-2019

ASE Department Faculty Committee, 2009-present

ASE Orbits Area Faculty Committee, 2009-present

ASE Controls, Autonomy, and Robotics Area Faculty Committee, 2014-present

ASE Graduate Studies Committee, 2009-present

ECE Graduate Studies Committee, 2012-present

HONORS AND AWARDS

Institute of Navigation Kepler Award, 2023

IEEE Walter Fried Best Paper Award, 2023 (with student Zach Clements)

Distinguished Alumnus of the Year, Utah State University ECE Department, 2022

Royal Institute of Navigation Fellow, 2021

IEEE Walter Fried Best Paper Award, 2020 (with student Lakshay Narula)

Institute of Navigation Fellow, 2020

PECASE: Presidential Early Career Award for Scientists and Engineers (NSF nomination), 2019

Qualcomm Innovation Fellowship, 2017 (with students Lakshay Narula and Matthew Murrian)

National Science Foundation CAREER Award, 2015

Outstanding Faculty Award, Dept. of Aerospace Engineering and Engineering Mechanics, 2015

Institute of Navigation Thurlow Award, 2014

GPS World Magazine Leadership Award, 2012

University of Texas System Regents' Outstanding Teaching Award, 2012

Cockrell School Dean's Award for Outstanding Teaching by an Assistant Professor, 2012

IEEE Walter Fried Best Paper Award, 2012 (with student Jahshan Bhatti)

Best Student Paper Award, IEEE/ION PLANS Conference, 2012 (with student Ken Pesyna)

PUBLICATIONS

Refereed Journal Publications

J1 T. E. Humphreys, P. A. Iannucci, and Z. Komodromos, "Timing properties of the Starlink Ku-band downlink," *IEEE Transactions on Aerospace and Electronic Systems*, 2024. In preparation.

J2 A. M. Graff and T. E. Humphreys, "Purposeful co-design of OFDM signals for ranging and communications," *EURASIP Journal on Advances in Signal Processing*, 2023. Submitted for review. pdf

J3 D. Akos, J. Arribas, M. Z. H. Bhuiyan, P. Closas, F. Dovis, I. Fernandez-Hernandez, S. Gunawardena, T. Humphreys, M. Nicola, T. Pany, M. Psiaki, A. Rügamer, J. A. L. Salcedo, C. Fernández-Prades, Y.-J. Song, and J.-H. Won, "GNSS software defined radio: History, current developments, and standardization efforts," *Navigation, Journal of the Institute of Navigation*, 2023. To be published.

J4 T. E. Humphreys, P. A. Iannucci, Z. M. Komodromos, and A. M. Graff, "Signal structure of the Starlink Ku-band downlink," *IEEE Transactions on Aerospace and Electronic Systems*, pp. 1–16, 2023 pdf

J5 J. E. Yoder and T. E. Humphreys, "Low-cost inertial aiding for deep-urban tightly-coupled multi-antenna precise GNSS," *Navigation, Journal of the Institute of Navigation*, vol. 70, no. 1, 2023 pdf

J6 P. A. Iannucci and T. E. Humphreys, "Fused low-earth-orbit GNSS," *IEEE Transactions on Aerospace and Electronic Systems*, pp. 1–1, 2022 pdf

J7 L. Narula, P. A. Iannucci, and T. E. Humphreys, "All-weather sub-50-cm radar-inertial positioning," *Field Robotics*, vol. 2, pp. 525–556, 2022 pdf

J8 W. A. Lies, L. Narula, P. A. Iannucci, and T. E. Humphreys, "Long-range, low SWaP-C FMCW radar," *IEEE Journal of Selected Topics in Signal Processing*, pp. 1–1, 2021 pdf

J9 M. J. Murrian, L. Narula, P. A. Iannucci, S. Budzien, B. W. O'Hanlon, S. P. Powell, and T. E. Humphreys, "First results from three years of GNSS interference monitoring from low Earth orbit," *Navigation, Journal of the Institute of Navigation*, vol. 68, no. 4, pp. 673–685, 2021 pdf

J10 T. E. Humphreys, M. J. Murrian, and L. Narula, "Deep-urban unaided precise global navigation satellite system vehicle positioning," *IEEE Intelligent Transportation Systems Magazine*, vol. 12, no. 3, pp. 109–122, 2020 pdf

J11 L. Narula, J. M. Wooten, M. J. Murrian, D. M. LaChapelle, and T. E. Humphreys, "Accurate collaborative globally-referenced digital mapping with standard GNSS," *Sensors*, vol. 18, no. 8, 2018 pdf

J12 J. N. Gross, C. Kilic, and T. E. Humphreys, "Maximum-likelihood power-distortion monitoring for GNSS-signal authentication," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 55, no. 1, pp. 469–475, 2018 pdf

J13 G. N. Green and T. Humphreys, "Position-domain integrity analysis for generalized integer aperture bootstrapping," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 55, no. 2, pp. 734–746, 2018 pdf

J14 G. N. Green and T. E. Humphreys, "Data-driven generalized integer aperture bootstrapping for high-integrity positioning," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 55, no. 2, pp. 757–768, 2018 pdf

J15 L. Narula and T. E. Humphreys, "Requirements for secure clock synchronization," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, pp. 749–762, Aug. 2018 pdf

J16 K. D. Wesson, J. N. Gross, T. E. Humphreys, and B. L. Evans, "GNSS signal authentication via power and distortion monitoring," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 54, pp. 739–754, April 2018 pdf

J17 K. M. Pesyna, Jr., T. Novlan, C. Zhang, R. W. Heath, Jr., and T. E. Humphreys, "Exploiting antenna motion for faster initialization of centimeter-accurate GNSS positioning with low-cost antennas," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 3, Aug. 2017 pdf

J18 J. Bhatti and T. E. Humphreys, "Hostile control of ships via false GPS signals: Demonstration and detection," *Navigation*, vol. 64, no. 1, pp. 51–66, 2017 pdf

J19 M. L. Psiaki and T. E. Humphreys, "GNSS spoofing and detection," *Proceedings of the IEEE*, vol. 104, no. 6, pp. 1258–1270, 2016 pdf

J20 Z. M. Kassas and T. E. Humphreys, "Receding horizon trajectory optimization in opportunistic navigation environments," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 51, pp. 866–877, April 2015 pdf

J21 Z. M. Kassas, A. Arapostathis, and T. E. Humphreys, "Greedy motion planning for simultaneous signal landscape mapping and receiver localization," *IEEE Journal of Selected Topics in Signal Processing*, vol. 9, pp. 247 – 258, March 2015 pdf

J22 C. R. Clauer, H. Kim, K. Deshpande, Z. Xu, D. Weimer, S. Musko, G. Crowley, C. Fish, R. Nealy, T. E. Humphreys, J. A. Bhatti, and A. J. Ridley, "Autonomous adaptive low-power instrument platform (AAL-PIP) for remote high latitude geospace data collection," *Geoscientific Instrumentation, Methods and Data Systems*, vol. 3, pp. 211–227, 2014 pdf

J23 A. J. Kerns, D. P. Shepard, J. A. Bhatti, and T. E. Humphreys, "Unmanned aircraft capture and control via GPS spoofing," *Journal of Field Robotics*, vol. 31, no. 4, pp. 617–636, 2014 pdf

J24 H. Kim, C. Clauer, K. Deshpande, M. Lessard, A. Weatherwax, G. S. Bust, G. Crowley, and T. E. Humphreys, "Ionospheric irregularities during a substorm event: Observations of ULF pulsations and GPS scintillations," *Journal of Atmospheric and Solar-Terrestrial Physics*, vol. 114, pp. 1–8, 2014 link

J25 K. M. Pesyna, Jr., Z. M. Kassas, R. W. Heath, Jr., and T. E. Humphreys, "A phase-reconstruction technique for low-power centimeter-accurate mobile positioning," *IEEE Transactions on Signal Processing*, vol. 62, pp. 2595–2610, May 2014 pdf

J26 Z. M. Kassas and T. E. Humphreys, "Observability analysis of collaborative opportunistic navigation with pseudorange measurements," *IEEE Transactions on Intelligent Transportation Systems*, vol. 15, pp. 260–273, Feb. 2014 pdf

J27 E. G. Lightsey, T. E. Humphreys, J. A. Bhatti, A. J. Joplin, B. W. O'Hanlon, and S. P. Powell, "Demonstration of a space capable miniature dual frequency GNSS receiver," *Navigation*, vol. 61, pp. 53–64, Mar. 2014 pdf

J28 T. E. Humphreys, "Detection strategy for cryptographic GNSS anti-spoofing," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 49, no. 2, pp. 1073–1090, 2013 pdf

J29 B. W. O'Hanlon, M. L. Psiaki, T. E. Humphreys, J. A. Bhatti, and D. P. Shepard, "Real-time GPS spoofing detection via correlation of encrypted signals," *Navigation, Journal of the Institute of Navigation*, vol. 60, no. 4, pp. 267–278, 2013 pdf

J30 M. Psiaki, B. O'Hanlon, J. Bhatti, D. Shepard, and T. Humphreys, "GPS spoofing detection via dual-receiver correlation of military signals," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 49, no. 4, pp. 2250–2267, 2013 pdf

J31 K. B. Deshpande, G. S. Bust, C. R. Clauer, H. Kim, J. E. Macon, T. E. Humphreys, J. A. Bhatti, S. B. Musko, G. Crowley, and A. T. Weatherwax, "Initial GPS scintillation results from CASES receiver at South Pole, Antarctica," *Radio Science*, vol. 47, no. 5, 2012 pdf

J32 D. P. Shepard, T. E. Humphreys, and A. A. Fansler, "Evaluation of the vulnerability of phasor measurement units to GPS spoofing attacks," *International Journal of Critical Infrastructure Protection*, vol. 5, no. 3-4, pp. 146–153, 2012 pdf

J33 K. D. Wesson, M. P. Rothlisberger, and T. E. Humphreys, "Practical cryptographic civil GPS signal authentication," *Navigation, Journal of the Institute of Navigation*, vol. 59, no. 3, pp. 177–193, 2012 pdf

J34 T. E. Humphreys, M. L. Psiaki, and P. M. Kintner, "Modeling the effects of ionospheric scintillation on GPS carrier phase tracking," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 46, pp. 1624–1637, Oct. 2010 pdf

J35 T. E. Humphreys, M. L. Psiaki, B. M. Ledvina, A. P. Cerruti, and P. M. Kintner, Jr., "A data-driven testbed for evaluating GPS carrier tracking loops in ionospheric scintillation," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 46, pp. 1609–1623, Oct. 2010 pdf

J36 T. E. Humphreys, M. L. Psiaki, J. C. Hinks, B. O'Hanlon, and P. M. Kintner, Jr., "Simulating ionosphere-induced scintillation for testing GPS receiver phase tracking loops," *IEEE Journal of Selected Topics in Signal Processing*, vol. 3, pp. 707–715, Aug. 2009 pdf

J37 T. E. Humphreys, M. C. Kelley, N. Huber, and P. M. Kintner, "The semidiurnal variation in GPS-derived zenith neutral delay," *Geophysical Research Letters*, vol. 32, no. 24, 2005 pdf

J38 T. E. Humphreys, M. Psiaki, E. Klatt, S. Powell, and P. M. Kintner, Jr., "Magnetometer-based attitude and rate estimation for a spacecraft with wire booms," *Journal of Guidance, Control, and Dynamics*, vol. 28, pp. 584–593, July–Aug. 2005 pdf

Refereed Conference Proceedings

C1 T. Haydon and T. E. Humphreys, "Trusted inertial terrain-aided navigation (TITAN)," in *Proceedings of the ION GNSS+ Meeting*, 2023 pdf

C2 W. Qin, Z. M. Komodromos, and T. E. Humphreys, "An agile, portable antenna system for LEO megaconstellation-based PNT," in *Proceedings of the ION GNSS+ Meeting*, 2023 pdf

C3 Z. M. Komodromos, W. Qin, and T. E. Humphreys, "Signal simulator for Starlink Ku-Band downlink," in *Proceedings of the ION GNSS+ Meeting*, 2023 pdf

C4 A. M. Graff and T. E. Humphreys, "Signal identification and entrainment for practical FMCW radar spoofing attacks," in *Proceedings of the IEEE 97th Vehicular Technology Conference (VTC2023-Spring)*, (Florence, IT), 2023 pdf

C5 R. M. Tenny, L. C. Sun, A. Duru, and T. E. Humphreys, "Robust absolute headset tracking for extended reality," in *Proceedings of the IEEE/ION PLANS Meeting*, (Monterey, CA), 2023 pdf

C6 Z. Clements, P. Ellis, and T. E. Humphreys, "Dual-satellite geolocation of terrestrial GNSS jammers from low Earth orbit," in *Proceedings of the IEEE/ION PLANS Meeting*, (Monterey, CA), pp. 458–469, 2023 pdf
Walter Fried overall best paper award.

C7 R. Tenny and T. E. Humphreys, "Robust navigation for urban air mobility via tight coupling of GNSS with terrestrial radionavigation and inertial sensing," in *Proceedings of the ION GNSS+ Meeting*, (Denver, CO), 2022 pdf
Best presentation award in session of 8 papers.

C8 Z. Clements, P. Ellis, M. L. Psiaki, and T. E. Humphreys, "Geolocation of terrestrial GNSS spoofing signals from low Earth orbit," in *Proceedings of the ION GNSS+ Meeting*, (Denver, CO), pp. 3418–3431, 2022 pdf

C9 D. Akos, J. Arribas, M. Z. H. Bhuiyan, P. Closas, F. Dovis, I. Fernandez-Hernandez, S. Gunawardena, T. Humphreys, M. Nicola, T. Pany, M. Psiaki, A. Rügamer, J. A. L. Salcedo, C. Fernández-Prades, Y.-J. Song, and J.-H. Won, "GNSS software defined radio history, current developments, and standardization efforts," in *Proceedings of the ION GNSS+ Meeting*, (Denver, CO), 2022 pdf

C10 H. A. Nichols, M. J. Murrian, and T. E. Humphreys, "Software-defined GNSS is ready for launch," in *Proceedings of the ION GNSS+ Meeting*, (Denver, CO), 2022 pdf

C11 Z. Clements, J. E. Yoder, and T. E. Humphreys, "Carrier-phase and IMU based GNSS spoofing detection for ground vehicles," in *Proceedings of the ION International Technical Meeting*, (Long Beach, CA), pp. 83–95, 2022 pdf

C12 D. M. LaChapelle, L. Narula, and T. E. Humphreys, "Orbital war driving: Assessing transient GPS interference from LEO," in *Proceedings of the ION GNSS+ Meeting*, (St. Louis, MO), 2021 pdf

C13 Z. Clements, P. A. Iannucci, T. E. Humphreys, and T. Pany, "Optimized bit-packing for bit-wise software-defined GNSS radio," in *Proceedings of the ION GNSS+ Meeting*, (St. Louis, MO), pp. 3749–3771, 2021 pdf

C14 A. M. Graff, W. N. Blount, P. A. Iannucci, J. G. Andrews, and T. E. Humphreys, "Analysis of OFDM signals for ranging and communications," in *Proceedings of the ION GNSS+ Meeting*, (St. Louis, MO), 2021 pdf

C15 R. X. Kor, P. A. Iannucci, and T. E. Humphreys, "Autonomous signal-situational awareness in a terrestrial radionavigation system," in *2021 International Conference on Intelligent Transportation Systems (ITSC)*, IEEE, 2021 pdf

C16 T. E. Humphreys, R. X. T. Kor, and P. A. Iannucci, "Open-world virtual reality headset tracking," in *Proceedings of the ION GNSS+ Meeting*, (Online), 2020 pdf

C17 J. E. Yoder, P. A. Iannucci, L. Narula, and T. E. Humphreys, "Multi-antenna vision-and-inertial-aided CDGNSS for micro aerial vehicle pose estimation," in *Proceedings of the ION GNSS+ Meeting*, (Online), pp. 2281–2298, 2020 pdf
Best presentation award in session of 8 papers.

C18 R. X. Kor, P. A. Iannucci, L. Narula, and T. E. Humphreys, "A proposal for securing terrestrial radio-navigation systems," in *Proceedings of the ION GNSS+ Meeting*, (Online), 2020 pdf

C19 L. Narula, P. A. Iannucci, and T. E. Humphreys, "Automotive-radar-based 50-cm urban positioning," in *Proceedings of the IEEE/ION PLANS Meeting*, 2020 pdf
Walter Fried overall best paper award.

C20 L. Narula, D. M. LaChapelle, M. J. Murrian, J. M. Wooten, T. E. Humphreys, J.-B. Lacambre, E. de Toldi, and G. Morvant, "TEX-CUP: The University of Texas Challenge for Urban Positioning," in *Proceedings of the IEEE/ION PLANS Meeting*, 2020 pdf

C21 P. A. Iannucci and T. E. Humphreys, "Economical fused LEO GNSS," in *Proceedings of the IEEE/ION PLANS Meeting*, 2020 pdf

C22 P. A. Iannucci, L. Narula, and T. E. Humphreys, "Cross-modal localization: Using automotive radar for absolute geolocation within a map produced with visible-light imagery," in *Proceedings of the IEEE/ION PLANS Meeting*, 2020 pdf

C23 W. A. Lies, L. Narula, P. A. Iannucci, and T. E. Humphreys, "Low SWaP-C radar for urban air mobility," in *Proceedings of the IEEE/ION PLANS Meeting*, 2020 pdf

C24 N. Montalbano and T. Humphreys, "Intercepting unmanned aerial vehicle swarms with neural-network-aided game-theoretic target assignment," in *Proceedings of the IEEE/ION PLANS Meeting*, 2020 pdf

C25 D. LaChapelle, T. E. Humphreys, L. Narula, P. A. Iannucci, and E. Moradi-Pari, "Automotive collision risk estimation under cooperative sensing," in *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing*, (Barcelona, Spain), 2020 pdf

C26 M. J. Murrian, L. Narula, and T. E. Humphreys, "Characterizing terrestrial GNSS interference from low earth orbit," in *Proceedings of the ION GNSS+ Meeting*, Institute of Navigation, Oct. 2019 pdf
Best presentation award in session of 8 papers.

C27 L. Narula, M. J. Murrian, and T. E. Humphreys, "Accuracy limits for globally-referenced digital mapping using standard GNSS," in *2018 21st International Conference on Intelligent Transportation Systems (ITSC)*, pp. 3075–3082, IEEE, 2018 pdf

C28 N. Montalbano and T. E. Humphreys, "A comparison of methods for online lever arm estimation in GPS/INS integration," in *2018 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, pp. 680–687, April 2018 pdf

C29 T. E. Humphreys, M. Murrian, and L. Narula, "Low-cost precise vehicular positioning in urban environments," in *2018 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, pp. 456–471, April 2018 pdf

C30 J. Gross and T. E. Humphreys, "GNSS spoofing, jamming, and multipath interference classification using a maximum-likelihood multi-tap multipath estimator," *Proceedings of the ION International Technical Meeting*, Jan. 2017 pdf

C31 L. Narula and T. E. Humphreys, "Requirements for secure wireless time transfer," in *Proceedings of the IEEE/ION PLANS Meeting*, (Savannah, GA), 2016 pdf

C32 G. N. Green, M. King, and T. E. Humphreys, "Data-driven generalized integer aperture bootstrapping for real-time high integrity applications," in *Proceedings of the IEEE/ION PLANS Meeting*, (Savannah, GA), 2016 pdf

C33 M. Murrian, C. Gonzalez, T. E. Humphreys, and T. D. Novlan, "A dense reference network for mass-market centimeter-accurate positioning," in *Proceedings of the IEEE/ION PLANS Meeting*, (Savannah, GA), 2016 pdf

C34 T. E. Humphreys, M. Murrian, K. M. Pesyna, Jr., F. van Diggelen, and S. Podshivalov, "On the feasibility of centimeter-accurate positioning via a smartphone's antenna and GNSS chip," in *Proceedings of the IEEE/ION PLANS Meeting*, (Savannah, GA), 2016 pdf

C35 G. N. Green, M. King, and T. E. Humphreys, "Fault free integrity of mid-level voting for triplex differential GPS solutions," in *Proceedings of the ION GNSS+ Meeting*, (Tampa, FL), 2015 pdf
Best presentation award in session of 8 papers.

C36 M. L. Psiaki, B. W. O'Hanlon, S. P. Powell, J. A. Bhatti, K. D. Wesson, T. E. Humphreys, and A. Schofield, "GNSS spoofing detection using two-antenna differential carrier phase," in *Proceedings of the ION GNSS+ Meeting*, (Tampa, FL), Institute of Navigation, 2014 pdf

C37 Z. M. Kassas, V. Ghadiok, and T. E. Humphreys, "Adaptive estimation of signals of opportunity," in *Proceedings of the ION GNSS+ Meeting*, 2014 pdf

C38 K. M. Pesyna, Jr., R. W. Heath, Jr., and T. E. Humphreys, "Centimeter positioning with a smartphone-quality GNSS antenna," in *Proceedings of the ION GNSS+ Meeting*, 2014 pdf

C39 T. E. Humphreys, D. P. Shepard, J. A. Bhatti, and K. D. Wesson, "A testbed for developing and evaluating GNSS signal authentication techniques," in *Proceedings of the International Symposium on Certification of GNSS Systems and Services (CERGAL)*, (Dresden, Germany), July 2014 pdf

C40 D. P. Shepard and T. E. Humphreys, "High-precision globally-referenced position and attitude via a fusion of visual SLAM, carrier-phase-based GPS, and inertial measurements," in *Proceedings of the IEEE/ION PLANS Meeting*, May 2014 pdf

C41 A. J. Kerns, K. D. Wesson, and T. E. Humphreys, "A blueprint for civil GPS navigation message authentication," in *Proceedings of the IEEE/ION PLANS Meeting*, May 2014 pdf

C42 K. D. Wesson, B. L. Evans, and T. E. Humphreys, "A probabilistic framework for Global Navigation Satellite System signal timing assurance," in *Proceedings of Asilomar Conference on Signals, Systems, and Computers*, (Pacific Grove, CA), 2013 pdf

C43 K. D. Wesson, B. L. Evans, and T. E. Humphreys, "A combined symmetric difference and power monitoring GNSS anti-spoofing technique," in *IEEE Global Conference on Signal and Information Processing*, 2013 pdf

C44 Z. M. Kassas, J. A. Bhatti, and T. E. Humphreys, "A graphical approach to GPS software-defined receiver implementation," in *Proceedings of IEEE Global Conference on Signal and Information Processing*, December 2013 pdf

C45 Z. Kassas and T. E. Humphreys, "The price of anarchy in active signal landscape map building," in *Proceedings of IEEE Global Conference on Signal and Information Processing*, December 2013 pdf

C46 Z. Kassas, J. A. Bhatti, and T. E. Humphreys, "Receding horizon trajectory optimization for simultaneous signal landscape mapping and receiver localization," in *Proceedings of the ION GNSS+ Meeting*, September 2013 pdf

C47 K. M. Pesyna, Jr., R. W. Heath, Jr., and T. E. Humphreys, "Precision limits of low-energy GNSS receivers," in *Proceedings of the ION GNSS+ Meeting*, (Nashville, Tennessee), Institute of Navigation, 2013 pdf
Best presentation award in session of 8 papers.

C48 Z. M. Kassas and T. E. Humphreys, "Motion planning for optimal information gathering in opportunistic navigation systems," in *AIAA Guidance, Navigation, and Control Conference (GNC'13)*, (Boston, MA), pp. 4551–4565, Aug. 2013 pdf

C49 K. D. Wesson, T. E. Humphreys, and B. L. Evans, "Position paper: Secure time transfer for CPS," in *NSF/NSA National Workshop on The New Clockwork for Time-Critical Systems*, 2012 pdf

C50 D. P. Shepard, K. M. Pesyna, Jr., and T. E. Humphreys, "Precise augmented reality enabled by carrier-phase differential GPS," in *Proceedings of the ION GNSS Meeting*, (Nashville, Tennessee), Institute of Navigation, 2012 pdf

C51 D. P. Shepard, J. A. Bhatti, T. E. Humphreys, and A. A. Fansler, "Evaluation of smart grid and civilian UAV vulnerability to GPS spoofing attacks," in *Proceedings of the ION GNSS Meeting*, 2012 pdf

C52 Z. M. Kassas and T. E. Humphreys, "Observability and estimability of collaborative opportunistic navigation with pseudorange measurements," in *Proceedings of the ION GNSS Meeting*, (Nashville, Tennessee), Institute of Navigation, 2012 pdf

C53 T. E. Humphreys, J. A. Bhatti, D. P. Shepard, and K. D. Wesson, "The Texas Spoofing Test Battery: Toward a standard for evaluating GNSS signal authentication techniques," in *Proceedings of the ION GNSS Meeting*, 2012 pdf

C54 Z. M. Kassas and T. E. Humphreys, "Observability analysis of opportunistic navigation with pseudorange measurements," in *Proceedings of AIAA Guidance, Navigation, and Control Conference*, Aug. 2012 pdf

C55 J. A. Bhatti, T. E. Humphreys, and B. M. Ledvina, "Development and demonstration of a TDOA-based GNSS interference signal localization system," in *Proceedings of the IEEE/ION PLANS Meeting*, pp. 1209–1220, April 2012 pdf
Walter Fried overall best paper award.

C56 K. M. Pesyna, Jr., Z. M. Kassas, and T. E. Humphreys, "Constructing a continuous phase time history from TDMA signals for opportunistic navigation," in *Proceedings of the IEEE/ION PLANS Meeting*, pp. 1209–1220, April 2012 pdf
Conference best student paper award.

C57 D. Shepard, T. E. Humphreys, and A. Fansler, "Evaulation of the vulnerability of Phasor Measurement Units to GPS spoofing," in *Sixth Annual IFIP WG 11.10 International Conference on Critical Infrastructure Protection*, (Washington, DC), Mar. 2012 pdf

C58 A. J. Joplin, E. G. Lightsey, and T. E. Humphreys, "Development and testing of a minaturized, dual-frequency GPS receiver for space applications," in *Proceedings of the ION International Technical Meeting*, (Long Beach, CA), Jan. 2012 pdf

C59 K. D. Wesson, M. P. Rothlisberger, and T. E. Humphreys, "A proposed navigation message authentication implementation for civil GPS anti-spoofing," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), Institute of Navigation, 2011 pdf

C60 D. Shepard and T. E. Humphreys, "Characterization of receiver response to a spoofing attack," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), Institute of Navigation, 2011 pdf

C61 R. Mitch, R. Dougherty, M. Psiaki, S. Powell, B. O'Hanlon, J. Bhatti, and T. Humphreys, "Signal characteristics of civil GPS jammers," in *Proceedings of the ION GNSS Meeting*, 2011 pdf
Best presentation award in session of 8 papers.

C62 M. L. Psiaki, B. W. O'Hanlon, J. A. Bhatti, and T. E. Humphreys, "Civilian GPS spoofing detection based on dual-receiver correlation of military signals," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), Institute of Navigation, 2011 pdf

C63 G. Crowley, G. S. Bust, A. Reynolds, I. Azeem, R. Wilder, B. W. O'Hanlon, M. L. Psiaki, S. Powell, T. E. Humphreys, and J. A. Bhatti, "CASES: A novel low-cost ground-based dual-frequency GPS software receiver and space weather monitor," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), Institute of Navigation, 2011 pdf

C64 B. O'Hanlon, M. Psiaki, S. Powell, J. Bhatti, T. E. Humphreys, G. Crowley, and G. Bust, "CASES: A smart, compact GPS software receiver for space weather monitoring," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), pp. 2745–2753, Institute of Navigation, 2011 pdf

C65 K. D. Wesson, D. P. Shepard, J. A. Bhatti, and T. E. Humphreys, "An evaluation of the vestigial signal defense for civil GPS anti-spoofing," in *Proceedings of the ION GNSS Meeting*, (Portland, OR), 2011 pdf
Best presentation award in session of 8 papers.

C66 K. M. Pesyna Jr., Z. M. Kassas, J. A. Bhatti, and T. E. Humphreys, "Tightly-coupled opportunistic navigation for deep urban and indoor positioning," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), Institute of Navigation, 2011 pdf

C67 K. M. Pesyna, Jr., K. D. Wesson, R. W. Heath, Jr., and T. E. Humphreys, "Extending the reach of GPS-assisted femtocell synchronization and localization through tightly-coupled opportunistic navigation," in *IEEE GLOBECOM Workshop*, 2011 pdf

C68 B. O'Hanlon, J. Bhatti, T. E. Humphreys, and M. Psiaki, "Real-time spoofing detection in a narrowband civil GPS receiver," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), Institute of Navigation, 2010 pdf

C69 T. E. Humphreys, J. Bhatti, and B. Ledvina, "The GPS Assimilator: a method for upgrading existing GPS user equipment to improve accuracy, robustness, and resistance to spoofing," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), Institute of Navigation, 2010 pdf

C70 K. D. Wesson, K. M. Pesyna, Jr., J. A. Bhatti, and T. E. Humphreys, "Opportunistic frequency stability transfer for extending the coherence time of GNSS receiver clocks," in *Proceedings of the ION GNSS Meeting*, (Portland, Oregon), Institute of Navigation, 2010 pdf

C71 T. E. Humphreys, J. Bhatti, T. Pany, B. Ledvina, and B. O'Hanlon, "Exploiting multicore technology in software-defined GNSS receivers," in *Proceedings of the ION GNSS Meeting*, (Savannah, GA), pp. 326–338, Institute of Navigation, 2009 pdf

C72 B. W. O'Hanlon, M. L. Psiaki, P. M. Kintner, Jr., and T. E. Humphreys, "Development and field testing of a DSP-based dual-frequency software GPS receiver," in *Proceedings of the ION GNSS Meeting*, (Savannah, GA), Institute of Navigation, 2009 pdf

C73 T. E. Humphreys, B. M. Ledvina, M. L. Psiaki, B. W. O'Hanlon, and P. M. Kintner, Jr., "Assessing the spoofing threat: Development of a portable GPS civilian spoofer," in *Proceedings of the ION GNSS Meeting*, (Savannah, GA), Institute of Navigation, 2008 pdf

C74 J. C. Hinks, T. E. Humphreys, B. O'Hanlon, M. L. Psiaki, and P. M. Kintner, Jr., "Evaluating GPS receiver robustness to ionospheric scintillation," in *Proceedings of the ION GNSS Meeting*, (Savannah, GA), Institute of Navigation, 2008 pdf
Best presentation award in session of 8 papers.

C75 S. Mohiuddin, T. E. Humphreys, and M. L. Psiaki, "A technique for determining the carrier phase differences between independent GPS receivers during scintillation," *Proceedings of the ION GNSS Meeting*, 2007 pdf
Best presentation award in session of 8 papers.

C76 M. L. Psiaki, T. E. Humphreys, S. Mohiuddin, S. P. Powell, A. P. Cerruti, and P. M. Kintner, Jr., "Searching for Galileo," in *Proceedings of the ION GNSS Meeting*, (Fort Worth, TX), Institute of Navigation, 2006 pdf
Best presentation award in session of 8 papers.

C77 T. E. Humphreys, B. M. Ledvina, M. L. Psiaki, and P. M. Kintner, Jr., "GNSS receiver implementation on a DSP: Status, challenges, and prospects," in *Proceedings of the ION GNSS Meeting*, (Fort Worth, TX), pp. 2370–2382, Institute of Navigation, 2006 pdf
Best presentation award in session of 8 papers.

C78 T. E. Humphreys, M. L. Psiaki, B. M. Ledvina, and P. M. Kintner, Jr., "GPS carrier tracking loop performance in the presence of ionospheric scintillations," in *Proceedings of the ION GNSS Meeting*, (Long Beach, CA), Institute of Navigation, Sept. 2005 pdf
Best presentation award in session of 8 papers.

C79 T. E. Humphreys, B. M. Ledvina, M. L. Psiaki, and P. M. Kintner, "Analysis of ionospheric scintillations using wideband GPS L1 C/A signal data," in *Proceedings of the ION GNSS Meeting*, (Long Beach, California), pp. 399–407, Institute of Navigation, 2004 pdf

C80 T. E. Humphreys, M. Kelley, and P. M. Kintner, Jr., "GPS-based measurement of atmospheric tides," in *Proceedings of the ION GNSS Meeting*, (Long Beach, California), pp. 864–880, Institute of Navigation, 2004

C81 T. E. Humphreys, "Attitude determination for small satellites with modest pointing constraints," in *Proc. 2002 AIAA/USU Small Satellite Conference*, (Logan, Utah), 2002

Other Major Publications

M1 Z. Clements, P. B. Ellis, and T. E. Humphreys, "Pinpointing GNSS interference from low earth orbit," *Inside GNSS*, vol. 18, no. 5, pp. 42–55, 2023

M2 R. Tenny and T. E. Humphreys, "Robust navigation for urban air mobility," *Inside GNSS*, vol. 18, no. 2, pp. 47–47, 2023

M3 Z. Clements, J. E. Yoder, and T. E. Humphreys, "GNSS spoofing detection: An approach for ground vehicles using carrier-phase and inertial measurement data," *GPS World*, vol. 34, no. 2, pp. 36–41, 2023

M4 J. E. Yoder, P. A. Iannucci, L. Narula, and T. E. Humphreys, "Seeing and inertial integrating is believing: Multi-antenna vision-and-inertial-aided CDGNSS for micro aerial vehicle pose estimation," *Inside GNSS*, vol. 16, no. 3, pp. 46–53, 2021

M5 M. J. Murrian, L. Narula, T. E. Humphreys, B. W. O'Hanlon, and S. Budzien, "Characterizing GNSS interference from low-earth orbit," *Inside GNSS*, vol. 15, no. 1, pp. 54–59, 2020

M6 M. J. Murrian, C. W. Gonzalez, T. E. Humphreys, K. M. P. Jr., D. P. Shepard, and A. J. Kerns, "Low-cost precise positioning for automated vehicles," *GPS World*, vol. 27, pp. 32–39, September 2016

M7 M. L. Psiaki and T. E. Humphreys, "Attackers can spoof navigation signals without our knowledge. here's how to fight back GPS lies," *IEEE Spectrum*, vol. 53, pp. 26–53, August 2016

M8 T. E. Humphreys, "Statement on the security threat posed by unmanned aerial systems and possible countermeasures," *United States House of Representatives Committee on Homeland Security: Subcommittee on Oversight and Management Efficiency*, Mar. 2015

M9 K. M. Pesyna, Jr, R. W. Heath, Jr., and T. E. Humphreys, "Accuracy in the palm of your hand: Centimeter positioning with a smartphone-quality GNSS antenna," *GPS World*, vol. 26, pp. 16–31, Feb. 2015

M10 M. L. Psiaki, B. W. O'Hanlon, S. P. Powell, J. A. Bhatti, T. E. Humphreys, and A. Schofield, "GNSS lies, GNSS truth: Spoofing detection with two-antenna differential carrier phase," *GPS World*, vol. 25, pp. 36–44, Feb. 2014

M11 K. D. Wesson and T. E. Humphreys, "Hacking drones," *Scientific American*, vol. 309, no. 5, pp. 54–59, 2013

M12 D. P. Shepard, J. A. Bhatti, and T. E. Humphreys, "Drone hack: Spoofing attack demonstration on a civilian unmanned aerial vehicle," *GPS World*, Aug. 2012

M13 D. P. Shepard, T. E. Humphreys, and A. A. Fansler, "Going up against time: The power grid's vulnerability to GPS spoofing attacks," *GPS World*, Aug. 2012

M14 T. E. Humphreys, "Statement on privacy issues related to the domestic use of unmanned aerial vehicles," *United States House of Representatives Committee on the Judiciary: Subcommittee on Crime, Terrorism, and Homeland Security: Field Forum*, Oct. 2012

M15 T. E. Humphreys, "Statement on the vulnerability of civil unmanned aerial vehicles and other systems to civil GPS spoofing," *United States House of Representatives Committee on Homeland Security: Subcommittee on Oversight, Investigations, and Management*, July 2012

M16 T. E. Humphreys, "The GPS dot and its discontents: Privacy vs. GNSS integrity," *Inside GNSS*, vol. 7, Mar./Apr. 2012

M17 R. Mitch, R. Dougherty, M. Psiaki, S. Powell, B. O'Hanlon, J. Bhatti, and T. E. Humphreys, "Know your enemy: Signal characteristics of civil GPS jammers," *GPS World*, Jan. 2012

M18 K. D. Wesson, D. P. Shepard, and T. E. Humphreys, "Straight talk on anti-spoofing: Securing the future of PNT," *GPS World*, Jan. 2012

M19 T. E. Humphreys, J. Bhatti, and B. M. Ledvina, "The GPS Assimilator: Upgrading receivers via benign spoofing," *Inside GNSS*, vol. 5, pp. 50–58, June 2010

M20 P. M. Kintner, Jr., T. E. Humphreys, and J. Hinks, "GNSS and ionospheric scintillation: How to survive the next solar maximum," *Inside GNSS*, vol. 4, pp. 22–30, July 2009

M21 P. Y. Montgomery, T. E. Humphreys, and B. M. Ledvina, "A multi-antenna defense: Receiver-autonomous GPS spoofing detection," *Inside GNSS*, vol. 4, pp. 40–46, April 2009

M22 T. E. Humphreys, B. M. Ledvina, M. L. Psiaki, B. W. O'Hanlon, and P. M. Kintner, Jr., "Assessing the spoofing threat," *GPS World*, vol. 20, pp. 28–38, Jan. 2009

M23 T. E. Humphreys, *Modeling Ionospheric Scintillation and its Effects on GPS Carrier Tracking Loops and Two Other Applications of Modeling and Estimation.* PhD thesis, Cornell University, Ithaca, New York, 2008

M24 T. E. Humphreys, L. Young, and T. Pany, "Considerations for future IGS receivers," in *Position Paper of the 2008 IGS Workshop*, 2008

M25 M. L. Psiaki, T. E. Humphreys, S. Mohiuddin, S. P. Powell, A. P. Cerruti, and J. Paul M. Kintner, "Searching for Galileo: Reception and analysis of signals from GIOVE-A," *GPS World*, vol. 17, pp. 66–72, June 2006

M26 T. E. Humphreys, "Attitude determination for small satellites with modest pointing constraints," Master's thesis, Utah State University, Logan, Utah, 2003

Book Chapters

B1 M. L. Psiaki and T. E. Humphreys, *Position, Navigation, and Timing Technologies in the 21st Century: Integrated Satellite Navigation, Sensor Systems, and Civil Applications*, vol. 1, ch. Civilian GNSS Spoofing, Detection, and Recovery, pp. 655–680. Wiley-IEEE, 2020

B2 T. E. Humphreys, "Interference," in *Springer Handbook of Global Navigation Satellite Systems*, pp. 469–503, Springer International Publishing, 2017

INVITED PRESENTATIONS

P1 September 15, 2023, "Pseudoranges from Starlink," (panel presentation) ION GNSS+ Conference, Denver, CO.

P2 September 15, 2023, "Interference Patterns," (panel presentation) ION GNSS+ Conference, Denver, CO.

P3 June 29, 2023, "Starlink Signal Structure for Navigation," International Technical Symposium on Navigation and Timing, ENAC, Toulouse, France.

P4 June, 2023 "Extraterrestrial 6G," Keysight World Conference (virtual).

P5 April 18, 2023, "Low-Cost Inertial Aiding for Deep-Urban Tightly-Coupled Multi-Antenna Precise GNSS," Institute of Navigation Webinar. video

P6 April 13, 2023, "PNT from LEO Mega-Constellations," SatCom LEO and MEO Space Technical Information Meeting & Exchange, Naval Postgraduate School, Monterey, CA.

P7 March 17, 2023, "Spoofing the Signal: Pinpointing Pirates and Putin," Utah State University, Logan, UT.

P8 March 17, 2023, "The Final Frontier is the Next One: The Geopolitics of Low-Earth Orbit," Center for Anticipatory Intelligence, Utah State University, Logan, UT.

P9 March 15, 2023, "Reverse Engineering the Starlink Downlink," Munich Satellite Navigation Summit, Munich, Germany.

P10 August 11, 2022, "Rolling Sensor Calibration: Lessons from the Vehicle Sensorium Project," Vision-Based Navigation Workshop (virtual).

P11 May 4, 2022, "First Results from Three Years of GNSS Interference Monitoring from Low Earth Orbit," STEP PNT Situational Awareness Workshop, Washington, D.C.

P12 February 7, 2022, "First Results from Three Years of GNSS Interference Monitoring from Low Earth Orbit," Institute of Navigation Webinar (virtual). video

P13 August 24, 2021, "Resilient and Robust PNT," (plenary presentation) Joint Navigation Conference, Cincinnati, Ohio. slides

P14 June 24, 2021, "Global GNSS Interference Activity," International Technical Symposium on Navigation and Timing, ENAC, Toulouse, France (virtual). video

P15 February 26, 2021, "GNSS Spoofing, Jamming, and Tomfoolery," SpaceX Seminar (virtual).

P16 December 3, 2020, "Radionavigation and Wireless Communications," NSF SII Workshop on Challenges and Opportunities for 6G and Navigation (virtual).

P17 July 10, 2020, "Cooperative Sensing for Automated Vehicles," Semiautonomous Seminar, U.C. Berkeley. video

P18 March 10, 2020, "All-Weather 50-cm Localization for Self-Driving Cars," Hexagon/NovAtel, Calgary, Canada. video

P19 December 13, 2019, "Belt-and-Suspenders PNT for Automated Vehicles," Joby Aviation, Santa Cruz, CA.

P20 November 19, 2019, "GNSS Radio Frequency Interference Detection from Low Earth Orbit," (keynote presentation) Royal Institute of Navigation International Navigation Conference, Edinburgh, Scotland.

P21 November 18, 2019, "Adding Resilience: Secure Perception for Autonomous Systems," Royal Institute of Navigation International Navigation Conference, Edinburgh, Scotland.

P22 October 16, 2019, "Collaborative Sensing for Self-Driving Cars: Security Implications," NSF Assured Autonomy Workshop, Washington, D.C.

P23 September 19, 2019, "Belt-and-Suspenders PNT for Self-Driving Cars and Air Taxis," (panel presentation) ION GNSS+ Conference, Miami, FL.

P24 September 18, 2019, "Insights from Two Years of GNSS Interference Observations from Space," (panel presentation), ION GNSS+ Conference, Miami, FL.

P25 July 10, 2019, "Automated Vehicle Control based on Secure Collaborative Sensing," American Control Conference: NSF CAREER awardee session, Philadelphia, PA.

P26 June 6, 2019, "GNSS Radio Frequency Interference Detection from LEO," National Executive Committee on Position Navigation and Timing, Advisory Board Meeting, Washington, DC.

P27 March 22, 2019, "The GPS Interference and Countermeasures Arms Race," Utah State University ECE Seminar, Logan, UT.

P28 Januray 15, 2019, "Deep Urban Unaided Precise GNSS Vehicle Positioning," West Virginia University.

P29 October 17, 2018, "Deep Urban Unaided Precise GNSS Vehicle Positioning," Technical University of Munich (TUM), Munich, Germany.

P30 October 16, 2018, "Bracing for the Future: Next-Generation Demands in PNT," German Aerospace Center (DLR), Munich, Germany.

P31 April 20, 2018, "Collaborative All-Weather Sensing for Automated Vehicles," Engineering Seminar, UT San Antonio.

P32 March 15, 2018, "Collaborative All-Weather Sensing for Automated Vehicles," Engineering Seminar, Tufts University.

P33 March 2, 2018, "Bracing for the Future: Next-Generation Demands in PNT," Aero-Astro seminar, Stanford University.

P34 March 1, 2018, "Collaborative All-Weather Sensing for Automated Vehicles," Apple Inc., special projects division.

P35 Dec. 15, 2017, "Collaborative All-Weather Sensing for Automated Vehicles," Apple Inc.

P36 Nov. 14, 2017, "Robust and secure perception for automated vehicles," (keynote presentation) International Technical Symposium on Navigation and Timing, ENAC, Toulouse, France.

P37 Oct. 9, 2017, "Robust and secure perception for automated vehicles," Engineering Seminar, Virginia Tech.

P38 Sept. 28, 2017, "Counter UAV challenges: Is GNSS spoofing effective," (panel presentation) ION GNSS+ panel session on hostile MAV threats, detection and countermeasures , Portland, OR.

P39 Sept. 28, 2017, "Assured navigation and timing," (panel presentation) ION GNSS+ panel session on assured navigation and timing, Portland, OR.

P40 Sept. 27, 2017, "Robust precise location," (panel presentation) ION GNSS+ panel session on ubiquitous navigation, Portland, OR.

P41 Feb. 18, 2017, "Secure Perception for Automated Vehicles," Hack the Machine, Austin, TX.

P42 Feb. 7, 2017, "Protecting Drivers and their Data," Texas Transportation Forum, Austin, TX.

P43 Dec. 15, 2016, "Trusted Automated Vehicles," IEEE Computer Society and COMSOC/SP Joint Chapters Meeting, Austin, TX.

P44 Dec. 5, 2016, "Robust and secure perception for automated vehicles," Nokia Bell Laboratories.

P45 Dec. 4, 2016, "Robust and secure perception for automated vehicles," (keynote presentation) Globecom location workshop, Washington D.C.

P46 Nov. 14, 2016, "Robust and secure perception for automated vehicles," Engineering Seminar, UCLA.

P47 Sept. 26, 2016, "Robust and secure perception for automated vehicles," Engineering Seminar, UC Riverside.

P48 April 25, 2016, "Low-cost precise positioning for automated vehicles," Hyundai Distinguished Lecture, UC Berkeley.

P49 February 9, 2016, "Precise Positioning for the Mass Market," (keynote presentation) International GNSS Service Workshop, Sydney, Australia. video

P50 October 13, 2015, "Low-Cost Precise Positioning and Perception Security," Google[X], Mountain View, CA.

P51 October 29, 2015, "Low-Cost Centimeter-Accurate Mobile Positioning," (keynote presentation) Texas GIS Forum, Austin, TX.

P52 October 1, 2015, "Low-Cost Centimeter-Accurate Mobile Positioning," University of Minnesota Roadway Safety Institute, Minneapolis, MN. video

P53 July 15, 2015, "Secure Perception for Autonomous Systems," (keynote presentation) International GNSS Conference, Gold Coast, Australia.

P54 June 11, 2015, "Toughening Techniques for GPS Receivers: Navigation Message Authentication," National Executive Committee on Position Navigation and Timing, Advisory Board Meeting, Washington, DC.

P55 March 18, 2015, "Unmanned aerial system threats: Exploring security implications and mitigation technologies," U.S. House Subcommittee on Homeland Security Oversight hearing, Washington, DC. link

P56 February 25, 2015, "Navigation Under Threat," (keynote presentation) International Navigation Conference, Manchester, UK.

P57 November 21, 2014, "Drones: Myths, Facts, Hacks, and the Future," Hot Science Cool Talks, The University of Texas at Austin Environmental Science Institute. link

P58 April 15, 2014, "Secure Perception for Autonomous Systems," University of Texas Student Engineering Council Symposium, Austin, TX.

P59 March 13, 2014, "Secure Perception for Autonomous Systems," University of Illinois at Urbana-Champaign, Urbana, IL.

P60 March 6, 2014, "Secure Perception for Autonomous Systems, (keynote presentation) Cornell Sibley Graduate Research Symposium, Ithaca, NY.

P61 March 7, 2014, "Location Deception," SXSW Interactive, Austin, TX. audio

P62 November 14, 2013, "Secure PNT for Autonomous Systems," Stanford PNT Symposium, Stanford University, Stanford, CA.

P63 September 26, 2013, "Maritime Navigation Security," International Hydrographic Organisation & Professional Yachters Association Sea Changes Seminar, Monaco.

P64 March 8, 2013, "Extreme GPS," SXSW Interactive, Austin, TX.

P65 March 6, 2013, "Drones in the Classroom," SXSW Edu, Austin, TX.

P66 February 13, 2013, "Secure Navigation and Timing," (keynote presentation) Royal Institute of Navigation Interference Conference, Teddington, UK.

P67 February 12, 2013, "UAV Integration: Privacy and Security Hurdles," Royal Institute of Navigation UAV Conference, Teddington, UK.

P68 February 7, 2013, "GPS Vulnerabilities and Implications for Telecom," international webinar.

P69 December 5, 2012, "Navigation and Timing Security," U.S. Air Force GPS Directorate, Los Angeles, CA.

P70 November 15, 2012, "Future Directions in GNSS Research," international GPS World webinar.

P71 October 25, 2012, "Privacy Issues Related to the Domestic Use of Unmanned Aerial Vehicles," statement to the U.S. House Judiciary Subcommittee on Crime, Terrorism, and Homeland Security field forum, Houston, TX. pdf

P72 October 17, 2012, "Secure Civil Navigation and Timing," Sandia National Laboratory, Albuquerque, NM.

P73 September 17, 2012, "Receiver Certification for Hardening Against Spoofing," Civil GPS Service Interface Committee, Nashville, TN.

P74 September 10, 2012, "Secure Civil Navigation and Timing," Aerospace Corporation research seminar, virtual from Austin, TX.

P75 July 20, 2012, "Secure Civil Navigation and Timing," MITRE corporation and government customers, McLean, VA.

P76 July 19, 2012, "Radionavigation Robustness and Security," Office of Naval Research, Arlington, VA.

P77 July 19, 2012, "The vulnerability of civil unmanned aerial vehicles and other systems to civil GPS spoofing," U.S. House Subcommittee on Homeland Security Oversight hearing on drone security, Washington, DC. video, pdf

P78 June 6, 2012, "Privacy vs. GPS Integrity," Civil GPS Service Interface Committee (CGSIC) Meeting, Austin, TX.

P79 February 22, 2012, "PVT Security: Privacy and Trustworthiness," (keynote presentation) Royal Institute of Navigation Conference on GNSS Vulnerability: Present Dangers, Future Threats 2012, Teddington, UK.

P80 February 11, 2012, "How to fool a GPS," TEDxAustin, Austin, TX. video

P81 December 2, 2011, "Radionavigation Robustness and Security," Draper Laboratory, Cambridge, MA.

P82 December 1, 2011, "Cubesat-Sized Radio Occultation Experiments," Massachusetts Institute of Technology Aeronautics and Astronautics Department, Cambridge, MA.

P83 April 4, 2011, "State of the art and future trends in radionavigation," US Patent and Trademark Office, virtual presentation from Austin, TX.

P84 March 10, 2011, "Briefing to DHS and DOD on GPS Security and Integrity," Austin, TX.

P85 October 4, 2010, "Spoofing the timing signal: What else is vulnerable?" National Executive Committee on Position Navigation and Timing, Advisory Board Meeting, Washington, DC.

P86 June 25, 2010, "Advances in GNSS Equipment," 2010 IGS Workshop, International GNSS Service, Newcastle upon Tyne, UK.

P87 January 12, 2010, "Riding out the rough spots: Scintillation-robust GNSS carrier tracking," 2010 Air Force Orion Conference, Dayton, OH.

P88 March, 2009, "Assessing the GPS spoofing threat," Cornell University, Ithaca, NY.

P89 June 5, 2008, "Considerations for future IGS receivers," 2008 IGS Workshop, International GNSS Service, Miami Beach, FL.

P90 March, 2006, "The semidiurnal variation in GPS-derived zenith neutral delay," 2006 IGS Workshop, Darmstadt, Germany.

Patents

P1 W. J. Bencze, C. E. Cohen, B. T. Galusha, T. E. Humphreys, B. M. Ledvina, and M. L. Psiaki, "Practical method for upgrading existing GNSS user equipment with tightly integrated nav-com capability," July 12 2011. US Patent 7,978,130

P2 K. J. Hayworth, K. V. Shcheglov, T. E. Humphreys, and A. D. Challoner, "Electrostatic spring softening in redundant degree of freedom resonators," Nov. 30 2004. US Patent 6,823,734

Software

W1 T. E. Humphreys, J. A. Bhatti, D. P. Shepard, and M. Murrian, "General radionavigation interfusion device (GRID) software suite (GSS)," 2021. UTA Tech ID 7912 HUM

W2 T. E. Humphreys, J. A. Bhatti, D. P. Shepard, and M. Murrian, "General radionavigation interfusion device (GRID) software suite (GSS)," 2019. UTA Tech ID 7417 HUM

W3 T. E. Humphreys, J. A. Bhatti, D. P. Shepard, A. J. Kerns, and K. M. P. Jr., "The precise positioning engine," 2019. UTA Tech ID 7245 HUM

W4 T. E. Humphreys, J. A. Bhatti, and D. P. Shepard, "Radionavigation security testbed software," 2012. UTA Tech ID 6199 HUM

W5 T. E. Humphreys, J. A. Bhatti, and D. P. Shepard, "General radionavigation interfusion device (GRID) software suite (GSS)," 2010. UTA Tech ID 5900 HUM

RESEARCH TOPICS

Application of optimal detection and estimation techniques to problems in secure, collaborative, and high-integrity perception, with an emphasis on positioning, navigation, and timing.

CONSULTING

U.S. Department of Homeland Security Risk Assessment, March - September, 2011 Coherent Navigation, August 2009 - August 2012

PH.D. SUPERVISION COMPLETED

1. L. Narula, *Towards Secure and Robust PNT for Automated Systems.* PhD thesis, The University of Texas at Austin, Dec. 2020. Current position: Senior Engineer at Google StreetView. **2021 Bradford Parkinson Best Dissertation Award from the Institute of Navigation.**
2. G. N. Green, *Advanced Techniques for Safety-of-Life Carrier Phase Differential GNSS Positioning with Applications to Triplex Architectures.* PhD thesis, The University of Texas at Austin, Dec. 2017. Current position: Senior engineer at CTSi and **Adjunct Professor, Electrical Engineering, LeTourneau University**
3. J. Bhatti, *Sensor Deception Detection and Radio-Frequency Emitter Localization.* PhD thesis, The University of Texas at Austin, Aug. 2015. Current position: Navigation software lead at Joby Aviation.

4. K. M. Pesyna, Jr., *Advanced Techniques for Centimeter-Accurate GNSS Positioning on Low-Cost Mobile Platforms*. PhD thesis, The University of Texas at Austin, Dec. 2015. Primary supervisor: Humphreys; Co-Supervisor: Robert Heath (ECE). **2015 Marconi Society Young Scholar Award.** Current position: Senior engineer at Apple, Inc.
5. Z. M. Kassas, *Analysis and Synthesis of Collaborative Opportunistic Navigation Systems*. PhD thesis, The University of Texas at Austin, May 2014. Primary supervisor: Humphreys; Co-supervisor: Aristotle Arapostathis (ECE). Current position: **Associate Professor, Mechanical and Aerospace Engineering, U.C. Irvine.**
6. K. Wesson, *Secure Navigation And Timing Without Local Storage Of Secret Keys*. PhD thesis, The University of Texas at Austin, May 2014. Primary supervisor: Humphreys; Co-supervisor: Brian Evans (ECE). Current position: Regulatory Engineer at SpaceX.

M.S. SUPERVISION COMPLETED

1. Hailey Nichols, H. A. Nichols, M. J. Murrian, and T. E. Humphreys, "Software-defined GNSS is ready for launch," in *Proceedings of the ION GNSS+ Meeting*, (Denver, CO), 2022
2. J. E. Yoder, "Low-cost inertial aiding for deep-urban tightly-coupled multi-antenna precise GNSS," Master's thesis, The University of Texas at Austin, 2021
3. R. X. T. Kor, "A comprehensive proposal for securing radionavigation systems," Master's thesis, The University of Texas at Austin, 2021
4. W. Lies, "Extending the range of low SWaP-C FMCW radar," Master's thesis, The University of Texas at Austin, 2021
5. Tucker Haydon, 2019
6. Connor Brashar, 2018
7. Greg Anders, 2017
8. Andrew Kerns, 2015
9. D. Shepard, "Fusion of carrier-phase differential GPS, bundle-adjustment-based visual slam, and inertial navigation for precisely and globally-registered augmented reality," Master's thesis, The University of Texas at Austin, May 2013.
10. M. W. Bright, "GPS L2C signal survey and the development of the emergent MATLAB L2C (EMAL2) receiver," Master's report, The University of Texas at Austin, Aug. 2012. Primary supervisor: Humphreys; Co-supervisor: Bob Schutz.
11. A. J. Joplin, "Development and testing of a miniaturized, dual-frequency, software-defined GPS receiver for space applications," Master's thesis, The University of Texas at Austin, Dec. 2011. Primary supervisor: Glenn Lightsey; Co-supervisor: Humphreys.

PH.D. SUPERVISION IN PROGRESS

1. Nick Montalbano
2. Daniel Lachapelle
3. Andrew Graff
4. Zach Clements
5. Zacharias Komodromos
6. Robert Tenny
7. Wenkai Qin
8. Alperen Duru

POST-DOCTORAL FELLOW SUPERVISION

1. Peter Iannucci, 2019-2022

OTHER STUDENT RESEARCH COMMITTEES (Current)

Ph.D. Defense Committees - 3
M.S. Committees - 1

OTHER RESEARCH SUPERVISION

Ph.D. Qualifying Committees

Kien Trung Truong, Kumar Appaiah, Thomas Novlan, Vidur Bhargava, Ehab Hussein Hassan, Yousof Mortazavi, Drew Jones, Jaegan Ko, Jing Lin, Chao Jia, Yezhou Wang

M.S. Report Committees

Constance McDaniel Wyman

Undergraduate Honors Projects

Daniel Shepard, Shubhodeep Mukherji, Siddarth Kaki, Tung To, James Yoder

Other Undergraduate Research Supervised

Vatsa Gandhi, Saura Elghonimi, Vish Selvakumar, Shaurya Gupta, Luciana Schement, Paige Whittington, Anum Aslam, Andrew Higdon, Sydney Norrell, Reese Shetrone, Zachary Tschirhart